

Financial Deserts in Latin America: A Subnational Atlas of Financial Exclusion

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Abstract

We construct the first subnational comparative database of financial access in Latin America, covering over 8,500 municipalities across Peru, Brazil, and Colombia from 2002 to 2025. We define “financial deserts” as municipalities with zero bank offices and document three findings. First, financial deserts are pervasive: 72.6% of Peruvian districts, 46.1% of Brazilian municipalities, and 25.1% of Colombian municipalities lack any bank office as of 2024. Second, deserts are extraordinarily persistent: 96% of Peruvian districts that were deserts in 2012 remain deserts in 2024. Third, all three countries are experiencing a *reversal* in physical financial infrastructure—bank offices have declined 15–24% from their peaks—suggesting that digitalization is closing existing offices rather than opening new ones in underserved areas. The atlas is publicly available as a contribution to the measurement of territorial inequality in the region.

Keywords: financial inclusion, financial deserts, Latin America, territorial inequality, bank branches

JEL Codes: G21, O16, R12

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1 Introduction

Financial inclusion is widely recognized as a driver of economic development and a dimension of inequality in its own right demirguckunt2018. Yet standard measures of financial access—such as the World Bank’s Global Findex or the IMF’s Financial Access Survey—operate at the national level, obscuring enormous subnational variation. A country may report 70% account ownership nationally while entire regions lack any financial infrastructure.

This paper introduces the concept of **financial deserts**: subnational units with zero bank offices. Borrowing from the “food deserts” literature in public health, the term captures territories where the complete absence of physical financial infrastructure creates a qualitatively different environment for economic activity. We construct the first comparative database of financial deserts across three Latin American countries—Peru, Brazil, and Colombia—using administrative data from their respective financial regulators.

Our contribution is threefold. First, we document the magnitude of financial deserts: nearly half of all subnational units across the three countries have zero bank offices. Second, we show that deserts are persistent structural traps, not transitional states. Third, we uncover a striking reversal: all three countries have experienced declining bank offices since their respective peaks in 2013–2019, with reductions of 15–24%. Rather than closing financial deserts, the expansion of digital finance appears to be closing offices in already-served areas.

The atlas is designed as a public good. The database, replication code, and an interactive visualization are available at <https://josemendozas.github.io/findedeserts/>.

2 Data and Methodology

We use administrative records from three financial regulators. For **Peru**, the Superintendencia de Banca, Seguros y AFP (SBS) provides the number of offices by type of institution per district, annually from 2002 to 2025. Institution types include commercial banks (Banca Múltiple), municipal savings banks (Cajas Municipales), rural savings banks, EDPYMES, and finance companies. For **Brazil**, the Banco Central do Brasil (BCB) publishes ESTBAN (Estatística Bancária por Município): a regulatory filing that records which bank agencies report deposits in each municipality each month. We use December snapshots from 2002 to 2024, accessed via basedosdados.org. Crucially, ESTBAN captures both openings and closures—an agency that stops reporting has effectively closed. For **Colombia**, the Superintendencia Financiera (SFC) reports the number of offices per entity per municipality, quarterly from 2008 to 2025, available through the national open data portal (datos.gov.co).

We define a **financial desert** as a subnational unit with zero bank offices in a given year. This “Layer 1” definition captures access to full financial services—credit origination, savings products, insurance, and advisory—which require physical presence and are not fully substitutable by digital channels or correspondent agents.

Population data come from national censuses: INEI 2017 (Peru), IBGE 2022 (Brazil), and DANE 2018 (Colombia). Peru’s panel is balanced (1,891 districts $\times$ 24 years). Brazil’s panel is balanced (5,570 municipalities $\times$ 23 years). Colombia’s panel uses the full universe of 1,121 municipalities for all years; municipalities absent from SFC filings in a given year are coded as having zero offices.

3 The Magnitude of Financial Deserts

Table 1 presents the cross-country comparison for 2024.

Table 1: Financial Deserts in Latin America, 2024

Country	Units	With Office	Deserts	% Desert	Offices
Peru	1,891	518	1,373	72.6%	3,986
Brazil	5,570	3,004	2,566	46.1%	8,311
Colombia	1,121	840	281	25.1%	5,425
Total	8,582	4,362	4,220	49.2%	17,722

Nearly half of all subnational units across the three countries—over 4,200 districts and municipalities—have zero bank offices. Peru’s rate of 72.6% reflects both the granularity of its administrative divisions (median district population: 3,090) and the concentration of banking infrastructure in coastal cities. Brazil’s 46.1% covers a far larger population base. Colombia, with a denser banking network, still has one in four municipalities without a bank office.

Figure 1 shows the evolution of desert rates over time.

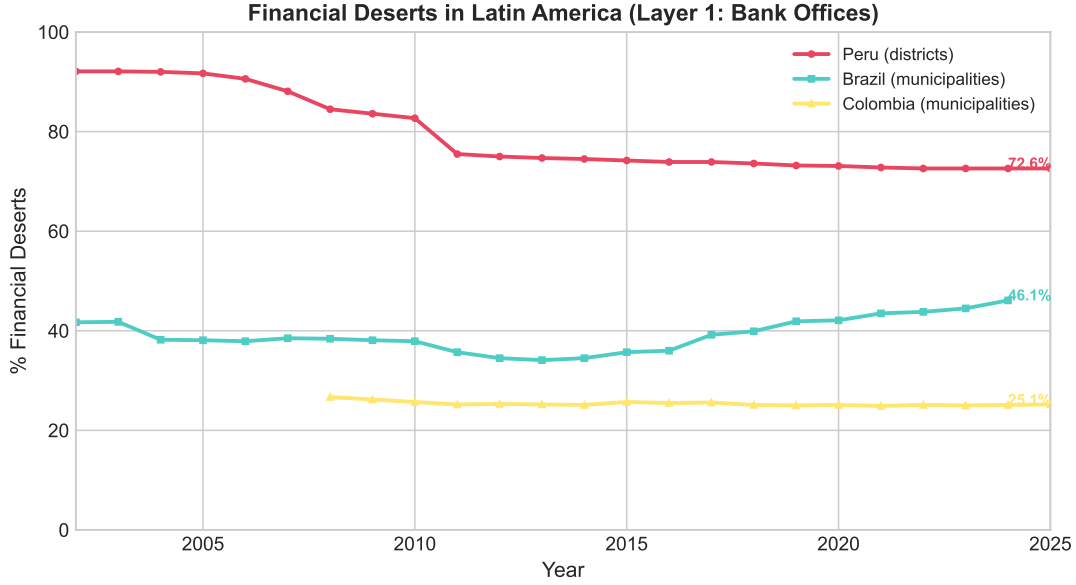


Figure 1: Percentage of subnational units classified as financial deserts, 2002–2025. Peru and Colombia show stable desert rates. Brazil’s rate declined from 2002 to 2013 but has been *increasing* since, driven by office closures.

The most notable pattern in Figure 1 is Brazil’s U-shaped trajectory. Brazilian financial deserts declined from 42% in 2002 to a minimum of 34% in 2013, then *reversed*, climbing back to 46% by 2024. Peru shows a more monotonic decline from 92% in 2002 to 73% by 2012, then stagnation. Colombia has remained stable at approximately 25% throughout the observed period.

4 Persistence

Financial deserts are not transitional states. Table 2 shows transition matrices for each country over the longest available window.

Table 2: Persistence of Financial Desert Status

Country	Period	Desert→Desert	Desert→Access	Access→Desert	Persistence
Peru	2012–2024	1,366	53	7	96.3%
Brazil	2002–2024	1,959	365	607	84.3%
Colombia	2008–2024	265	34	16	88.6%

In Peru, 96.3% of districts that were deserts in 2012 remain deserts 12 years later—only 53 districts gained a bank office. In Brazil, persistence is lower (84.3%) but the direction of change is alarming: 607 municipalities *lost* their last bank agency, exceeding the 365 that gained one. Brazil is the only country in our sample experiencing net *desertification*. Figure 2 illustrates the annual flow.

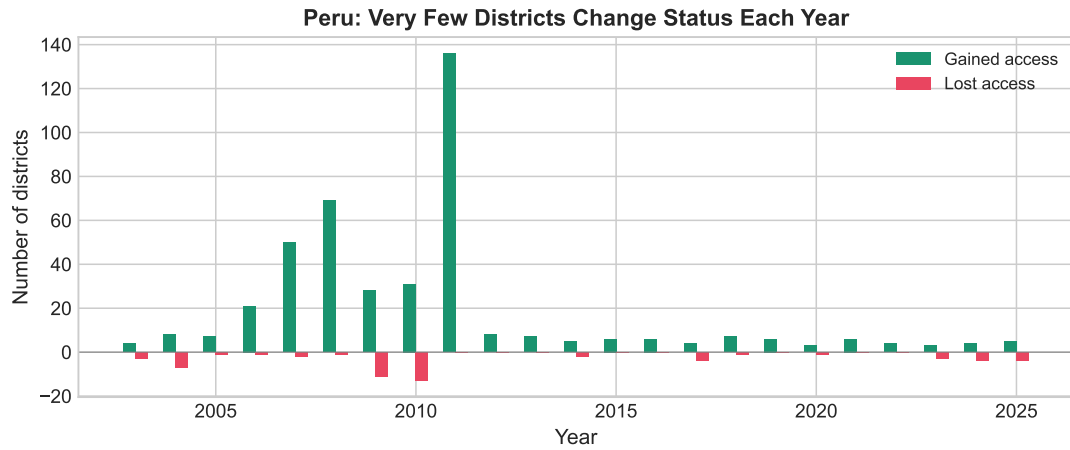


Figure 2: Annual transitions in and out of financial desert status, Peru 2003–2025. The number of districts changing status each year is minimal—typically fewer than 30—illustrating the structural nature of deserts.

5 The Office Reversal

Perhaps the most striking finding is that all three countries are experiencing a contraction in physical banking infrastructure. Figure 3 shows the trend.

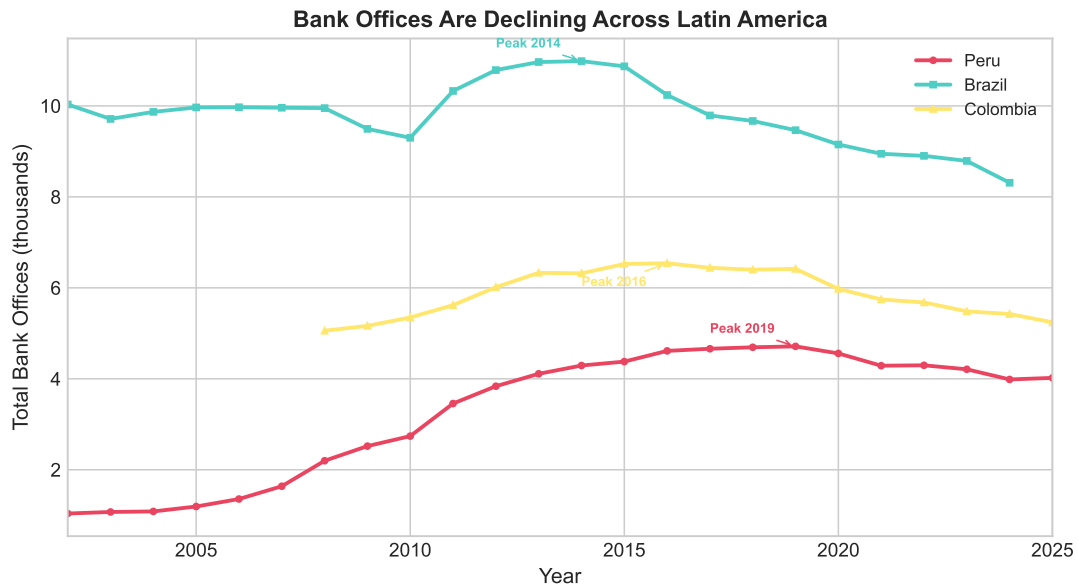


Figure 3: Total bank offices by country, 2002–2025. All three countries show declining office counts from their respective peaks (Peru 2019, Brazil 2014, Colombia 2016).

Table 3: Bank Office Decline from Peak

Country	Peak Year	Peak Offices	2024 Offices	Decline
Peru	2019	4,713	3,986	−15.4%
Brazil	2014	10,985	8,311	−24.3%
Colombia	2016	6,540	5,425	−17.1%

Brazil’s decline is the most severe: nearly one in four bank agencies has closed since 2014, coinciding with the rapid adoption of PIX (instant payments, launched 2020) and mobile banking. Peru’s decline accelerated after 2019, alongside the expansion of Yape and Plin. Colombia shows a similar pattern with a 17% reduction since 2016.

Crucially, these closures are concentrated in municipalities that *already had offices*—not in financial deserts. The net effect is a narrowing of the served territory: more municipalities losing their last office, rather than an expansion into underserved areas. In Brazil, 607 municipalities lost their last bank agency between 2002 and 2024.

6 Discussion

Three implications emerge from these findings.

First, financial deserts represent a spatial dimension of inequality that standard income-based measures miss. Two households with identical income face radically different economic opportunities depending on whether they live in a municipality with bank offices. Without physical infrastructure, residents lack access to credit origination, formal savings, insurance, and financial advisory—services that digital channels do not fully substitute, particularly for complex financial products.

Second, the simultaneous expansion of digital finance and contraction of physical banking raises a fundamental question: are digital channels *complementing* physical infrastructure or *substituting* it? Our evidence is consistent with substitution in already-served areas. The municipalities gaining digital access are predominantly those that already had physical banking, while financial deserts remain underserved by either channel. This “digital displacement” hypothesis warrants further investigation, particularly as regulators across the region promote digital finance as a solution to exclusion.

Third, our Layer 1 definition—zero bank offices—is deliberately conservative. Many deserts have correspondent agents (Peru has over 166,000 agent points reaching 99% of districts) or digital wallets. Future versions of the atlas will incorporate these additional layers, enabling researchers to measure the gap between transactional inclusion (can you withdraw cash?) and intermediation inclusion (can you obtain a loan for your business?).

The atlas has several limitations. The administrative level differs across countries: Peru’s districts are more granular than Brazilian or Colombian municipalities, mechanically inflating Peru’s desert rate. Additionally, the population-weighted desert rate—what share of people live in deserts—would provide a more policy-relevant measure than the unit-based rate we report here. We leave this extension for subsequent work.

7 Conclusion

We introduce the Financial Deserts Atlas, a subnational database of physical financial access covering Peru, Brazil, and Colombia from 2002 to 2025. Three facts emerge. Financial deserts are pervasive—nearly half of all subnational units lack any bank office. They are persistent—over 84–96% remain deserts over a decade. And physical banking infrastructure is contracting, not expanding, as digital finance grows.

These findings suggest that the financial inclusion agenda in Latin America may be solving the wrong problem: expanding digital transactions while the infrastructure for financial intermediation retreats. The atlas is publicly available at <https://josemendozas.github.io/findeserts/>, with replication code on GitHub. We invite researchers to use and extend this resource.

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